```
# Update system repositories
sudo apt update

# Install required system tools and libraries
sudo apt install -y python3-pip python3-serial python3-rpi.gpio i2c-tools

# Install Python modules using --break-system-packages (required for Raspberry Pi OS Bookworm)
pip3 install RPLCD adafruit-circuitpython-fingerprint mfrc522 --break-system-packages
```

Programme Number 01 (RFID)

```
from mfrc522 import SimpleMFRC522
import RPi.GPIO as GPIO

reader = SimpleMFRC522()

try:
    print("Place RFID card...")
    while True:
        uid, text = reader.read()
        print("RFID UID:", uid)
        print("Data:", text)
        break
finally:
    GPIO.cleanup()
```

Programme Number 02 (FINGERPRINT)

```
import serial
import adafruit_fingerprint

# UART configuration on Raspberry Pi GPIO 14/15
uart = serial.Serial("/dev/serial0", baudrate=57600, timeout=1)
finger = adafruit_fingerprint.Adafruit_Fingerprint(uart)

location = int(input("Enter fingerprint ID (1-127): "))

print("Place finger...")
while finger.get_image() != adafruit_fingerprint.OK:
    pass

if finger.image_2_tz(1) != adafruit_fingerprint.OK:
    print("Image conversion failed")
    exit()

print("Remove finger")
```

```
input("Press Enter after removing finger...")

print("Place same finger again...")
while finger.get_image() != adafruit_fingerprint.OK:
    pass

if finger.image_2_tz(2) != adafruit_fingerprint.OK:
    print("Image conversion failed")
    exit()

if finger.create_model() != adafruit_fingerprint.OK:
    print("Fingerprint mismatch")
    exit()

if finger.store_model(location) == adafruit_fingerprint.OK:
    print("Fingerprint enrolled successfully")
else:
    print("Enrollment failed")
```

Programme Number 03 (main Programme)

```
import RPi.GPIO as GPIO
import time
import csv
import os
import serial
from datetime import datetime
from RPLCD.i2c import CharLCD
from mfrc522 import SimpleMFRC522
import adafruit_fingerprint

# Pin Definitions (BCM numbering)
RELAY = 23
LED = 21

GPIO.setmode(GPIO.BCM)
GPIO.setup(RELAY, GPIO.OUT)
GPIO.setup(LED, GPIO.OUT)

# Active-LOW Relay: default HIGH (OFF)
GPIO.output(RELAY, GPIO.HIGH)
GPIO.output(LED, GPIO.LOW)

# Initialize I2C LCD (change 0x27 to 0x3F if i2cdetect reported 0x3F)
lcd = CharLCD(
    i2c_expander='PCF8574',
    address=0x27,
```

```
    port=1,
    cols=16,
    rows=2,
    charmap='A00',
    auto_linebreaks=True
)

reader = SimpleMFRC522()

# Replace with the decimal UID obtained from rfid_test.py
WHITELIST = [
    123456789,
    987654321
]

# Initialize Fingerprint Sensor
uart = serial.Serial("/dev/serial0", baudrate=57600, timeout=1)
finger = adafruit_fingerprint.Adafruit_Fingerprint(uart)

LOG_FILE = "access_log.csv"

def log_event(method, result):
    new_file = not os.path.exists(LOG_FILE)
    with open(LOG_FILE, "a", newline="") as file:
        writer = csv.writer(file)
        if new_file:
            writer.writerow(["Time", "Method", "Result"])
        writer.writerow([
            datetime.now().strftime("%Y-%m-%d %H:%M:%S"),
            method,
            result
        ])

def show_message(line1, line2=""):
    lcd.clear()
    lcd.write_string(line1[:16])
    if line2:
        lcd.cursor_pos = (1, 0)
        lcd.write_string(line2[:16])

def unlock(method):
    show_message("ACCESS GRANTED", method)
    GPIO.output(LED, GPIO.HIGH)
    GPIO.output(RELAY, GPIO.LOW)  # Active-LOW trigger
    log_event(method, "GRANTED")
    time.sleep(5)
    GPIO.output(RELAY, GPIO.HIGH) # Turn off relay
    GPIO.output(LED, GPIO.LOW)
```

```
def deny(method):
    show_message("ACCESS DENIED", method)
    GPIO.output(LED, GPIO.LOW)
    log_event(method, "DENIED")
    time.sleep(2)

def check_fingerprint():
    # Poll sensor for up to 3 seconds
    start = time.time()
    while time.time() - start < 3:
        result = finger.get_image()
        if result == adafruit_fingerprint.OK:
            break
        elif result == adafruit_fingerprint.NOFINGER:
            time.sleep(0.1)
            continue
        else:
            return False
    else:
        return False

    show_message("Processing...")
    if finger.image_2_tz(1) != adafruit_fingerprint.OK:
        return False
    if finger.finger_search() != adafruit_fingerprint.OK:
        return False
    return True

try:
    show_message("ACCESS CONTROL", "SYSTEM READY")
    time.sleep(2)

    while True:
        show_message("Scan RFID", "or Fingerprint")

        # Non-blocking RFID check
        card_id, text = reader.read_no_block()
        if card_id:
            print("RFID UID:", card_id)
            if card_id in WHITELIST:
                unlock("RFID")
            else:
                deny("RFID")
            time.sleep(1)
            continue

        # Check Fingerprint sensor
```

```
        if check_fingerprint():
            print("Fingerprint matched")
            unlock("FINGERPRINT")
            time.sleep(1)

except KeyboardInterrupt:
    print("\nProgram stopped")
finally:
    GPIO.output(RELAY, GPIO.HIGH)
    GPIO.output(LED, GPIO.LOW)
    GPIO.cleanup()
    lcd.clear()
```